RUNWAY STUDIOS

AI Growth OS — Google Ads Integration

Technical Design Document

Version	1.0
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Website	runwaystudios.co
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API Access Level	Basic Access (Requested)

1. Product Overview

Runway Studios AI Growth OS is a SaaS marketing intelligence platform built for Indian D2C brands and digital agencies. It aggregates advertising performance data from multiple platforms — Meta Ads, Google Ads, and YouTube — into a unified analytics dashboard, and applies AI (Claude by Anthropic) to generate cross-channel performance insights and prioritised optimisation recommendations.

The platform is hosted on Google Cloud Run (asia-south1) and serves clients through a web dashboard. All Google Ads data is accessed via the Google Ads API exclusively for read-only reporting purposes.

> *Core Principle: The Google Ads API is used solely for reading performance data. The application does not create, modify, pause, enable, or delete any campaigns, ad groups, ads, keywords, or any other Google Ads entity.*

2. System Architecture

The platform consists of four main layers:

- Frontend Dashboard — Next.js 14 web app served from Google Cloud Run
- Backend API — Python FastAPI service handling business logic and API orchestration
- Database — PostgreSQL on Google Cloud SQL for storing KPIs, tokens, and workspace config
- AI Engine — Anthropic Claude API for generating cross-channel recommendations

Component	Technology	Role	Location
Frontend Dashboard	Next.js 14, Tailwind CSS	Analytics UI for clients	Cloud Run (asia-south1)
Backend API	Python 3.11, FastAPI	Business logic, API orchestration	Cloud Run (asia-south1)
Database	PostgreSQL 14 (Cloud SQL)	KPI data, OAuth tokens, config	Cloud SQL (asia-south1)
AI Engine	Claude API (Anthropic)	Cross-channel analysis, recommendations	Anthropic API (external)

Component	Technology	Role	Location
Google Ads Connector	google-ads Python SDK v24	Fetches metrics via GAQL	Within backend service

3. Authentication & Authorisation

OAuth2 FlowOAuth2 Flow

The platform uses Google's standard OAuth2 authorisation code flow. No user passwords are stored at any point. Access is granted explicitly by the account owner through Google's consent screen.

1. User clicks "Connect Google Ads" in the Settings page
2. Platform redirects to accounts.google.com/oauth2/auth with requested scopes
3. User reviews and consents on Google's consent screen
4. Google returns authorisation code to our callback endpoint
5. Backend exchanges code for access_token + refresh_token
6. Tokens stored encrypted in Cloud SQL, scoped to the user's workspace_id
7. All subsequent Google Ads API calls use the stored refresh token

OAuth2 Scopes RequestedOAuth2 Scopes Requested

Scope	Purpose	Access Type
https://www.googleapis.com/auth/adwords	Read Google Ads performance data via GAQL	Read Only
https://www.googleapis.com/auth/youtube.readonly	Read YouTube channel stats and video metadata	Read Only
https://www.googleapis.com/auth/yt-analytics.readonly	Read YouTube Analytics (watch time, CTR)	Read Only

Token Storage & SecurityToken Storage & Security

- Access tokens and refresh tokens stored in encrypted columns in PostgreSQL (Cloud SQL, asia-south1)
- Tokens are workspace-scoped — each client's tokens are isolated by workspace_id
- Refresh tokens are used automatically when access tokens expire (1-hour standard expiry)
- Users can revoke access at any time via Google Account settings or the platform's Settings page
- Tokens are never logged, exposed in API responses, or sent to the frontend

4. Google Ads API Usage

All API calls use GAQL (Google Ads Query Language) SELECT queries only. No mutate operations are performed. The following table lists every report type accessed:

Report Type	GAQL Resource	Fields Retrieved	Dashboard Purpose
Campaign Performance	campaign	name, status, impressions, clicks, cost, conversions, conversion_value, date	KPI cards, spend charts, ROAS calculation

Report Type	GAQL Resource	Fields Retrieved	Dashboard Purpose
Keyword Performance	keyword_view	keyword_text, match_type, quality_score, impressions, clicks, cost, conversions	Keyword analysis, QS badges, wasted spend
Search Terms	search_term_view	search_term, campaign, ad_group, clicks, cost, conversions	Negative keyword recommendations
Geographic Performance	geographic_view	location_type, location_name, campaign, impressions, clicks, cost, conversions	City/region ROAS, expansion opportunities
Device Performance	campaign (device segment)	device, campaign, impressions, clicks, cost, conversions	Mobile vs desktop budget split
Time of Day	campaign (hour/day segment)	hour_of_day, day_of_week, cost, conversions	Ad scheduling heatmap
Auction Insights	auction_insight	competitor_domain, impression_share, overlap_rate, position_above_rate	Competitive position analysis
Ad Assets (RSA)	ad_group_ad	asset_text, asset_type, performance_label, impressions, clicks	Creative performance analysis
Account Discovery	CustomerService.listAccessibleCustomers	customer_id, descriptive_name	Multi-account agency management

No Mutate Operations: The application does not call any Google Ads API write methods. There are no calls to CampaignService.mutate, AdGroupService.mutate, BudgetService.mutate, KeywordService.mutate, or any other operation that modifies Google Ads data.

5. Data Flow & Storage

Ingestion PipelineIngestion Pipeline

1. User authenticates via OAuth2 and grants read access to their Google Ads account
2. FastAPI backend calls Google Ads API using the stored refresh token
3. Performance metrics retrieved for the requested date range via GAQL SELECT queries
4. Data normalised and stored in the kpi_hourly table in PostgreSQL, keyed by workspace_id, platform, entity_level, and timestamp
5. Next.js dashboard reads this data via internal REST API endpoints (authenticated)
6. Claude AI analyses aggregated data and returns text recommendations (not persisted)

Data Retention & IsolationData Retention & Isolation

Data Type	Storage Location	Isolation Key	Retention Policy
OAuth tokens	Cloud SQL (encrypted columns)	workspace_id	Until revoked or account deleted

Data Type	Storage Location	Isolation Key	Retention Policy
Campaign KPI data	Cloud SQL · kpi_hourly table	workspace_id	Duration of active account
Keyword / search term data	Cloud SQL · kpi_hourly table	workspace_id	Duration of active account
AI-generated recommendations	Not stored — generated on-demand per session	N/A	Not persisted

6. Multi-Account (Agency) Support

The platform supports agencies managing multiple client Google Ads accounts. Each client account maps to a separate Workspace. The implementation follows Google's recommended Manager Account (MCC) model:

- A single OAuth2 token from the agency's Google account (with MCC access) covers multiple client accounts
- CustomerService.listAccessibleCustomers called once to discover available customer IDs
- Each GAQL query specifies the target login-customer-id header to scope results to the correct client
- Client data strictly isolated by workspace_id — no cross-client data leakage is architecturally possible

Scenario	Implementation
Brand managing own account	Direct OAuth2 with their Google account. Single customer_id used in all requests.
Agency managing client accounts	OAuth2 with agency MCC account. Per-client login-customer-id header in each GAQL request.

7. Security & Compliance

Security Control	Implementation	Status
Data encryption at rest	Google Cloud SQL with encryption enabled by default	Active
Data encryption in transit	TLS 1.2+ on all API endpoints — Cloud Run managed certificates	Active
OAuth token security	Tokens stored in encrypted DB columns, never exposed in logs or API responses	Active
API authentication	All internal endpoints require Bearer token — no unauthenticated access	Active
Data residency	All data stored in Google Cloud asia-south1 (Mumbai, India)	Active
No third-party data sharing	Google Ads data never shared with, sold to, or transferred to any third party	Active
Access revocation	Users can disconnect Google Ads at any time via Settings page or Google Account	Active
Minimal scope principle	Only read-only scopes requested — no write, manage, or admin scopes	Active

Security Control	Implementation	Status
Workspace isolation	All DB queries filter by workspace_id — tenant data fully isolated	Active

8. Google Ads API Policy Compliance

Policy Requirement	How Runway Studios Complies
Accurate application information	All information provided reflects the actual production system at runwaystudios.co
Data use limitation	Google Ads data used exclusively for analytics and AI recommendations within the platform, only for the data owner
No resale of data	Google Ads performance data is never sold, licensed, or transferred to any third party
No scraping or bulk export	Data retrieved for dashboard display and AI analysis only — no bulk exports or redistribution
User consent required	Each user explicitly grants access via Google's OAuth2 consent screen before any API call is made
Valid privacy policy	Published at runwaystudios.co/#privacy — includes specific Google Ads API data handling section
API contact email monitored	hello@runwaystudios.co — checked daily, response within 24 business hours
Read-only access	No mutate, create, or delete operations are performed on any Google Ads entity

9. Technology Stack Summary

Layer	Technology	Version / Details
Backend Language	Python	3.11+
Backend Framework	FastAPI + Uvicorn	Latest stable
Google Ads SDK	google-ads (Python client library)	24.x
Frontend Framework	Next.js (React)	14.2
Database	PostgreSQL (Google Cloud SQL)	14
Hosting	Google Cloud Run	asia-south1 (Mumbai)
AI / LLM	Claude API (Anthropic)	claude-sonnet-4-6
Authentication	Google OAuth2 + Clerk (dashboard users)	Standard OAuth2 flow
Domain	runwaystudios.co	Hosted on GitHub Pages + GoDaddy DNS

10. Contact & Support

Type	Details
Company Name	Runway Studios
Website	https://runwaystudios.co
API Contact Email	hello@runwaystudios.co
Support Phone	+91 88262 83840
Dashboard URL	https://dashboard-771420308292.asia-south1.run.app
Privacy Policy	https://runwaystudios.co/#privacy
Data Storage Location	Google Cloud asia-south1 (Mumbai, India)
Developer Token Location	Google Ads Manager Account — API Center